

Q #1

Multiple Choice Type

Award: 1

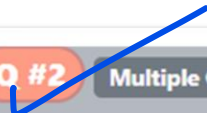
Penalty: 0.33

Machine Learning

Given data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$, we obtain $\hat{w}$, the parameters that minimize the training error cost for the linear regression model $y = w^T x$ learned from D .

Now consider a new dataset D_{new} , generated by duplicating the points in D and adding 10 points that lie exactly along the line $y = \hat{w}^T x$. Then the $\hat{w}_{\text{new}}$ learned from D_{new} is:

- A. $\hat{w}_{\text{new}} = \hat{w}$
- B. $\hat{w}_{\text{new}} = 10\hat{w}$
- C. $\hat{w}_{\text{new}} = \frac{\hat{w}}{10}$
- D. $\hat{w}_{\text{new}} \neq \hat{w}$

 Q #2

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

You are building a linear regression model to predict future stock value y using past monthly averages $(x_1, x_2, \dots, x_n)$.

Suppose your model has high test error. A colleague suggests adding new features based on differences between consecutive months:

$$(x_1 - x_2), (x_2 - x_3), \dots, (x_{n-1} - x_n).$$

Will adding these new features necessarily reduce the test error?

- A. Yes, because new features always increase the expressive power of linear regression.
- B. No, because these different features are just linear combinations of the original features, so they do not add new information.
- C. Yes, because different features capture temporal trends that the model cannot capture otherwise.
- D. It depends: the test error may decrease if regularization is applied, but without it the model remains the same.

Q #3

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Consider the following two regression models:

$$\text{Model 1: } \hat{y} = w^2x + wx$$

$$\text{Model 2: } \hat{y} = wx$$

Suppose we have limited training data available.

Which of the following statements is correct regarding their ability to fit the training data?

- A. Model 1 will fit the data better
- B. Model 2 will fit the data better
- C. Both will fit the data equally well
- D. It is impossible to determine

Q #4

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Let w^* be the solution obtained in standard least-squares linear regression using the dataset

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

Now, suppose we construct a new dataset by scaling all the input features by a constant factor c , i.e.,

$$D' = \{(cx_1, y_1), (cx_2, y_2), \dots, (cx_n, y_n)\}$$

Let the regression solution obtained on D' be denoted by $\tilde{w}$.

Which of the following correctly relates $\tilde{w}$ to the original solution w^* ?

A. $\tilde{w} = \frac{1}{c} w^*$

B. $\tilde{w} = c w^*$

C. $\tilde{w} = \frac{1}{c^2} w^*$

D. $\tilde{w} = c^2 w^*$

Q #5

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Suppose we have one qualitative variable that we convert to numerical values using one-hot encoding. We've shown the first four rows of the resulting design matrix below:

a	b	c
1	0	0
1	0	0
0	0	1
0	1	0

Say we train a linear model m_1 on these data. Then, we replace all of the 1 values in column **a** with 3's and all of the 1 values in column **b** with 2's and train a new linear model m_2 . Neither m_1 nor m_2 has an intercept term. On the training data, the average squared loss for m_1 will be _____ that of m_2 .

- A. greater than
- B. less than
- C. equal to
- D. impossible to tell

Q #7

Numerical Type

Award: 2

Penalty: 0

Machine Learning

Consider the following 1-D dataset:

$$(1, 2), (2, 4), (3, 6), (4, 6), (5, 6)$$

We use the hypothesis:

$$h(x) = 2x$$

Define the loss function $L(\hat{y}, y)$ as follows:

$$L(\hat{y}, y) = \begin{cases} 0 & \text{if } |\hat{y} - y| \leq 2 \\ |\hat{y} - y| - 2 & \text{if } |\hat{y} - y| > 2 \end{cases}$$

What is the average loss of this dataset under the above hypothesis?
(Answer up to two decimal places.)

Q #6

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

Fit a line $y = a + bx$ to the points $(0, 0), (1, 2), (2, 1), (3, 3)$ by minimizing

$$S(a, b) = \sum_{i=1}^4 \left(y^{(i)} - (a + bx^{(i)}) \right)^2$$

Using $\frac{\partial S}{\partial a} = 0$ and $\frac{\partial S}{\partial b} = 0$, which pair of linear equations in a, b do you obtain?

- A. $4a + 6b = 6$ and $6a + 14b = 13$
- B. $4a + 6b = 8$ and $6a + 14b = 15$
- C. $4a + 10b = 6$ and $6a + 18b = 13$
- D. $6a + 6b = 6$ and $6a + 12b = 13$

Q #8

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

In a weighted linear regression problem with dataset

$$(x_1, y_1) = (-1, 0), (x_2, y_2) = (0, 1), (x_3, y_3) = (1, 0),$$

and weighted loss function

$$L(w) = \frac{1}{3} \left[(y_1 - \hat{y}_1)^2 + 2(y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 \right],$$

we fit a predictor $\hat{y} = w_0 + w_1 x$.

Notice that the middle point $(0, 1)$ is given double the weight compared to the others.

Which of the following is correct about the optimal solution?

A. $w_0 = \frac{1}{2}, w_1 = 0$

B. $w_0 = 0, w_1 = 1$

C. $w_0 = 1, w_1 = 0$

D. $w_0 = \frac{1}{3}, w_1 = \frac{1}{3}$

Q #9

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

A regression model is tested on a dataset with **1,000** observations. After comparing the predictions $\hat{y}$ with the true values y , the following was observed:

- For **400** observations, the prediction error $(\hat{y} - y)$ was **+2** (the model predicted 2 more than the actual).
- For **300** observations, the prediction error $(\hat{y} - y)$ was **-1** (the model predicted 1 less than the actual).
- For the remaining **300** observations, the prediction error was **0** (perfect prediction).

The mean squared error (MSE) is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

What is the MSE of this model?

- A. 1.3
- B. 1.6
- C. 1.9
- D. 2.6

Q #10

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

A linear regression model is trained on a dataset of N points, giving parameters $\hat{w}$. Now a new dataset is formed in the following way:

- Each of the N original points is triplicated (so we now have $3N$ copies of the same data).
- Additionally, 20 new points are added, and all of these new points lie exactly on the line defined by the original model $\hat{y} = \hat{w}^T x$.

Let $\hat{w}_{\text{new}}$ be the parameters obtained from training linear regression on this new dataset.

Which statement is correct?

- A. $\hat{w}_{\text{new}} = \hat{w}$ (parameters remain the same)
- B. $\hat{w}_{\text{new}}$ will definitely change due to triplication
- C. $\hat{w}_{\text{new}}$ will change because of the 20 new points
- D. $\hat{w}_{\text{new}}$ cannot be determined without more information

Q #11

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

We use linear regression to model a dataset of size N .

A random subset of the data is chosen as the training set, and the remaining points are used as the test set.

Now suppose the training set size is gradually increased.

As the training set size grows, what is the expected behavior of the mean training error E_{train} and the mean testing error E_{test} ?

- A. E_{train} increases, E_{test} decreases
- B. E_{train} decreases, E_{test} increases
- C. E_{train} increases, E_{test} increases
- D. E_{train} decreases, E_{test} decreases

Q #12

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

Consider a simple linear regression model with an intercept:

$$\hat{y}_i = w_0 + w_1 x_i, \quad i = 1, 2, \dots, n$$

where the residuals are defined as

$$r_i = y_i - \hat{y}_i$$

Which of the following properties of the residuals always hold true for the least squares solution?

A. $\sum_{i=1}^n r_i = 0$

B. $\sum_{i=1}^n x_i r_i = 0$

C. Both A and B

D. Neither A nor B

Q #13

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

Emma is fitting a simple linear regression model *without an intercept*, of the form $\hat{y} = w_E x$. Her dataset is shown on the **left**. Liam records the *same outcomes* y but measures inputs in different units: each input is scaled by a constant $c > 0$. He then fits the model $\hat{y} = w_L z$. His dataset is shown on the **right**.

Emma's Dataset

Observation	x	y
1	1	2
2	2	4
3	3	6

Liam's Dataset

Observation	$z = c x$	y
1	$c \cdot 1$	2
2	$c \cdot 2$	4
3	$c \cdot 3$	6

Let the training errors obtained by Emma and Liam be denoted $\mathcal{L}_E$ and $\mathcal{L}_L$, respectively. Which of the following statements is correct?

A. $w_L = w_E$, and $\mathcal{L}_E = \mathcal{L}_L$.

☒ B. $w_L = \frac{1}{c} w_E$, and $\mathcal{L}_E = \mathcal{L}_L$.

C. $w_L = w_E$, but $\mathcal{L}_E \neq \mathcal{L}_L$.

D. $w_L = c w_E$, and $\mathcal{L}_E = \mathcal{L}_L$.

Q #14

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

In simple linear regression, suppose we introduce a rescaled feature $x' = 10x$. We then fit the following two models:

$$\text{Model A: } \hat{y} = \theta_0 + \theta_1 x$$

$$\text{Model B: } \hat{y} = \beta_0 + \beta_1 x'$$

Which of the following statements about the relationship between (θ_0, θ_1) and (β_0, β_1) is correct?

A. $\hat{\theta}_1 = \hat{\beta}_1$ and $\hat{\theta}_0 = \hat{\beta}_0$

B. $\hat{\beta}_1 = \frac{1}{10} \hat{\theta}_1$ and $\hat{\theta}_0 \neq \hat{\beta}_0$

C. $\hat{\beta}_1 = \frac{1}{10} \hat{\theta}_1$ and $\hat{\theta}_0 = \hat{\beta}_0$

D. $\hat{\theta}_1 = 10 \hat{\beta}_1$ and $\hat{\theta}_0 \neq \hat{\beta}_0$

Q #15

Multiple Select Type

Award: 2

Penalty: 0

Machine Learning

Suppose you have trained a polynomial model $y(x) = w_0 + w_1 x + \dots + w_M x^M$ via least-squares regression, and you find that it has a low training error but a high testing error. Which of the following is likely to reduce the testing error?

A. Increasing the number of training data points and re-training.

B. Decreasing M and re-training.

C. Dividing each of the learnt weights by 2.

D. Increasing M and re-training.

Your Answer:

Correct Answer: A;B

Not Attempted

Time taken: 00min 00sec

Discuss